Ponnamanda Manoj Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.80
Tirumala Junior College	Class XII (BIEAP)	2023	97.80%
Tirumala High Schooel	Class X (BSEAP)	2021	100.00%

PROJECTS

AI Travel Planner | *Self Project* **|** Google ADK, Python, MCP

- Built a **multi-agent AI travel planner** using Google ADK, orchestrating specialized agents for trip intake, flights, lodging, activities, budget reconciliation, and itinerary generation.
- Designed a **concurrent agent pipeline** with ParallelAgent and an iterative LoopAgent to research travel options in parallel and automatically resolve budget and scheduling conflicts.
- Integrated **Amadeus and Google Maps APIs** through tool-augmented LLM agents, with mock-backed fallbacks enabling deterministic testing and development without external API dependencies.
- Extended the system with **MCP-based logistics tools** and a read-only **BigQuery price-intelligence agent**, while implementing conditional callbacks and pytest-based behavioral evaluation for robust end-to-end workflows.

AI Research Navigator | *Self Project* **|** Python, LangGraph, Qdrant, FastEmbed

- **Built a citation-grounded Retrieval-Augmented Generation (RAG) system** over a curated corpus of **50 AI/ML documents**, implementing PDF/Markdown parsing, content-type-aware chunking, deterministic SHA-256 content hashing, and metadata-rich document representations for reproducible ingestion.
- **Implemented a hybrid retrieval pipeline** using **Qdrant, FastEmbed (BGE dense embeddings + BM25 sparse retrieval), and Reciprocal Rank Fusion (RRF)**, with server-side metadata filtering, deterministic query understanding, content-addressed document updates, and a cosine-similarity refusal gate for low-confidence queries.
- **Designed a LangGraph agent with 6 specialized routes**, combining deterministic routing with LLM fallback and multiple anti-hallucination gates for citation validation, source deduplication, low-confidence refusal, and graceful abstention; added a modular evaluation framework with routing, retrieval, faithfulness, and cost metrics.

Multi-Client Quiz System with LLM Integration | C++, Python, TCP/IP

Course Project under Prof. Kotaro Kataoka and Prof. Praveen Tammana

- Built a **concurrent TCP client-server quiz system in C++** with dynamic LLM-powered question generation and real-time leaderboard updates.
- Designed a **custom binary application-layer protocol** supporting keep-alives, stateful session management, and seamless integration with a Python/SSE web frontend.
- Optimized **LLM latency and network bandwidth** using TTL-based caching, reducing redundant upstream requests and improving performance under multi-client workloads.
- Automated **concurrent load testing** with Python-based client bots and profiled network performance using Wireshark, measuring throughput, packet behavior, and end-to-end latency.

Adaptive Latency-Aware Handshaking Protocol (ALHP) | C++17, Python, NS-3, UDP

Course Project under Prof. Kotaro Kataoka and Prof. Praveen Tammana

- Designed an adaptive **UDP transport protocol for short-lived IoT connections**, dynamically selecting 0-RTT Immediate, 0-RTT Token, or 1-RTT Full handshake modes using live RTT and packet-loss metrics.
- Built a **platform-agnostic C++17 protocol core** with IClock/ITransport abstractions, supporting NS-3 simulation, POSIX sockets, and unit-test environments.
- Implemented **RFC 6298-based RTT/RTO estimation**, sliding-window loss tracking, and confidence-aware mode selection with hysteresis and dwell-time controls.
- Developed **session-token and metric caching** for low-latency 0-RTT adaptation, and benchmarked performance against the ALHP-PROBE measure-before-send baseline.

SKILLS

Languages: C, C++, Python, SQL

Machine Learning: Deep Learning, Transformers, LLMs

RELEVANT COURSEWORK

Core Courses: Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering

AI/ML: Machine Learning, Deep Learning,Multi Agent Systems

MATHS: Linear Optimization, Convex Optimization, Probability and Random Variables

LEADERSHIP & ACADEMIC ROLES

Class Representative for CSE'27 Batch

Head of Events Diesta '25: Led a team in the planning and execution of technical and cultural events for Diesta, the annual college Sports and Cultural fest.